LAB 2: AGILE BACKLOG CREATION & SPRINT SIMULATION IN JIRA

Webhook Ingestion & Retry Mechanism Hub

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Problem Statement	#49 – Webhook Ingestion & Retry Mechanism Hub

1. Objective

Use Jira to turn the functional requirements from Lab 1 into Agile backlog items, organize them as Epics and User Stories, prioritize and estimate them using Story Points, simulate two sprints, and analyse sprint progress using the Burndown Chart.

2. Project Overview

The project is a resilient webhook gateway that receives third-party webhook events, logs payloads, forwards them to configured internal services, automatically retries failed deliveries using exponential backoff, and stores permanently failed events in a dead-letter queue.

3. Epics and User Stories

KAN-4 – Webhook Ingestion & Delivery

Description: Enable the system to receive inbound webhooks, log their payloads, and deliver them to configured internal target endpoints.

Issue Key	User Story	Priority	Story Points
KAN-7	Ingest and Deliver Webhook	High	8
KAN-10	Log Webhook Payload	High	3

KAN-5 – Retry & Failure Management

Description: Manage failed webhook deliveries through automatic retries with exponential backoff and dead-letter queue storage when all delivery attempts fail.

Issue Key	User Story	Priority	Story Points
KAN-11	Retry Failed Webhook Delivery	High	5
	Store Webhook in Dead-Letter Queue	High	3

KAN-6 – Configuration & Monitoring

Description: Allow integration developers to configure webhook endpoints and retry settings, and system operators to monitor delivery logs and dead-letter queue entries.

Issue Key	User Story	Priority	Story Points

	Configure Webhook Endpoint and Retry Settings	Medium	5
	Monitor Delivery Logs and Dead-Letter Queue	Medium	5

Total estimated backlog: 29 Story Points.

4. Story Point Estimation

Fibonacci-based Story Points were assigned to represent relative complexity, effort, and uncertainty. The estimates were used for sprint planning and Burndown Chart tracking.

User Story	Priority	Story Points
Ingest and Deliver Webhook	High	8
Log Webhook Payload	High	3
Retry Failed Webhook Delivery	High	5
Store Webhook in Dead-Letter Queue	High	3
Configure Webhook Endpoint and Retry Settings	Medium	5
Monitor Delivery Logs and Dead-Letter Queue	Medium	5

5. Sprint Planning and Simulation

Two one-week sprints were planned. Sprint 1 focused on the core ingestion, logging, and retry functionality. Sprint 2 covered dead-letter queue handling, configuration, and monitoring.

Sprint	Planned User Stories	Total Points
Sprint 1	Ingest and Deliver Webhook; Log Webhook Payload; Retry Failed Webhook Delivery	16
Sprint 2	Store Webhook in Dead-Letter Queue; Configure Webhook Endpoint and Retry Settings; Monitor Delivery Logs and Dead-Letter Queue	13

During the simulation, the User Stories were progressed through the Jira workflow: To Do → In Progress → Done. Both planned sprints were completed.

6. Jira Screenshots

Insert the following screenshots in this section. The Lab 2 handout requires screenshots of the Jira backlog with Epics and User Stories, Story Point assignments, the Sprint Board (Active Sprint view), and the Burndown Chart.

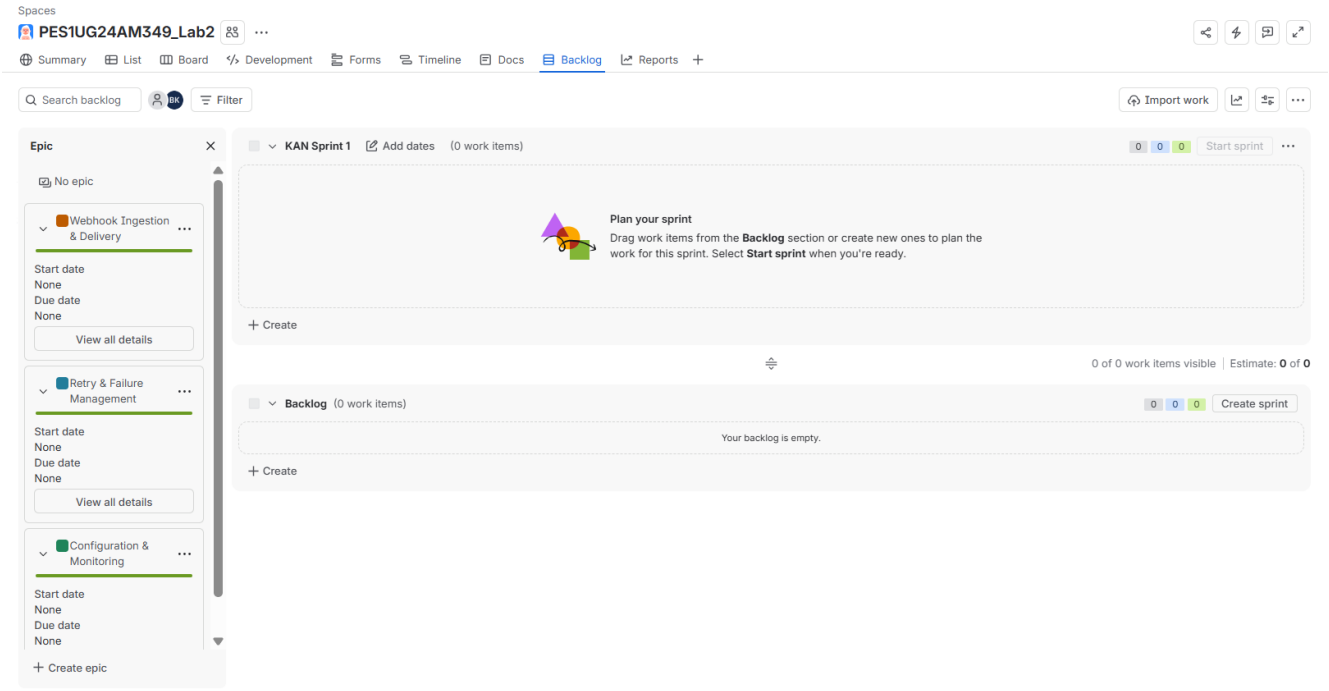


Figure 1: Jira Backlog showing Epics and User Stories

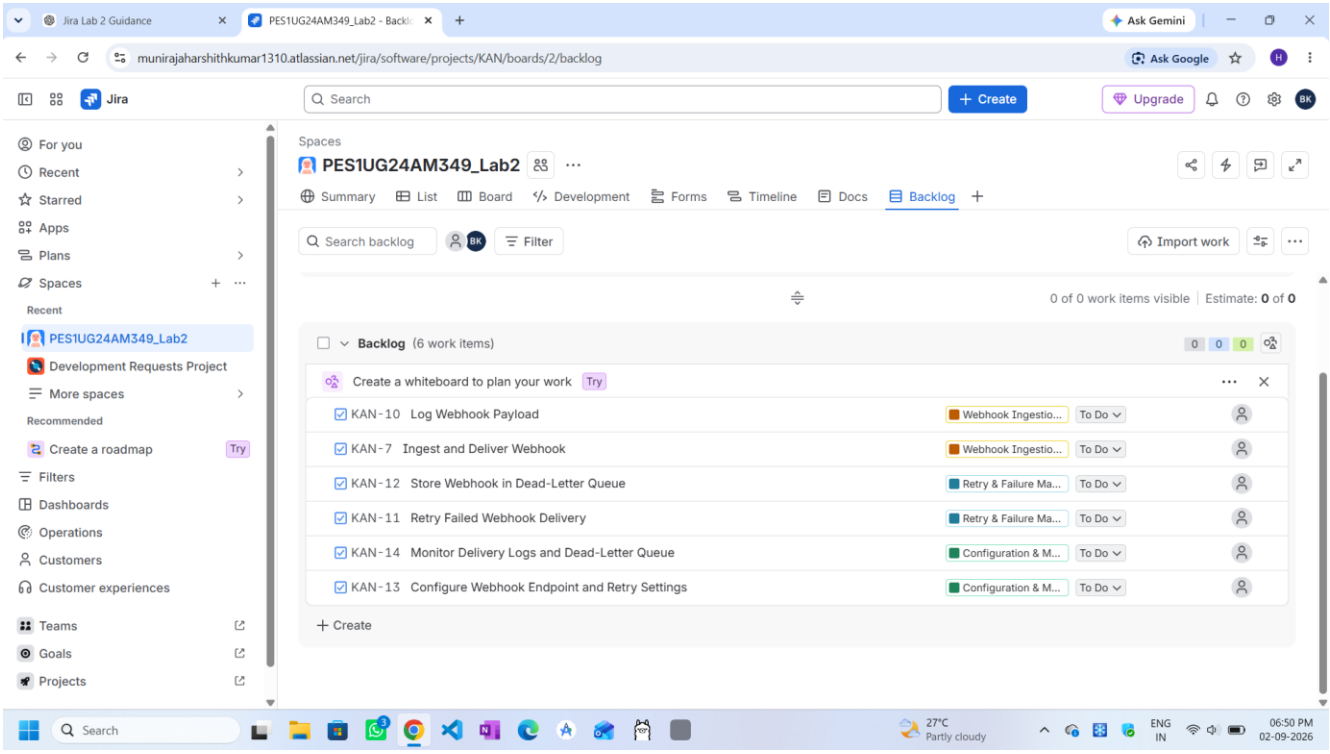


Figure 2: Jira Backlog / User Stories showing Story Point assignments

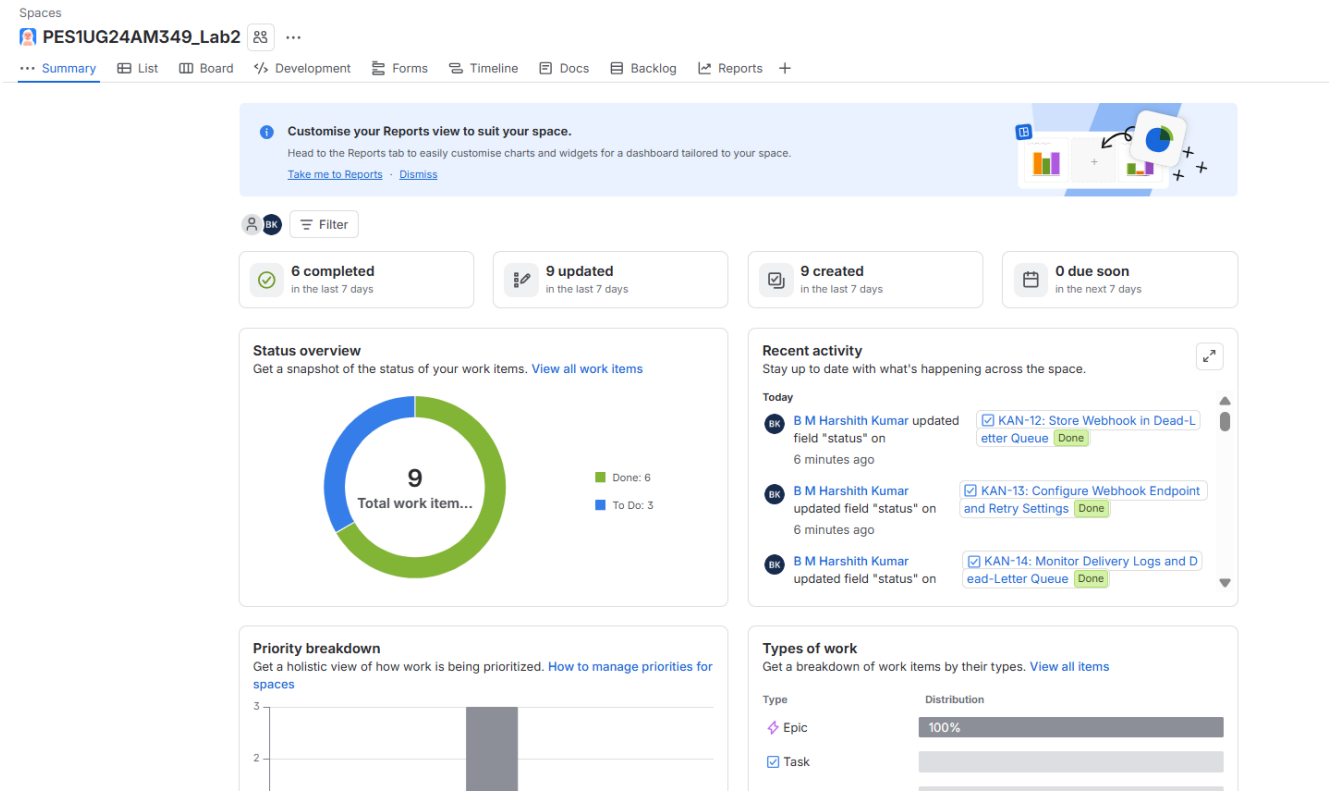


Figure 3: Active Sprint / Sprint Board showing task progress

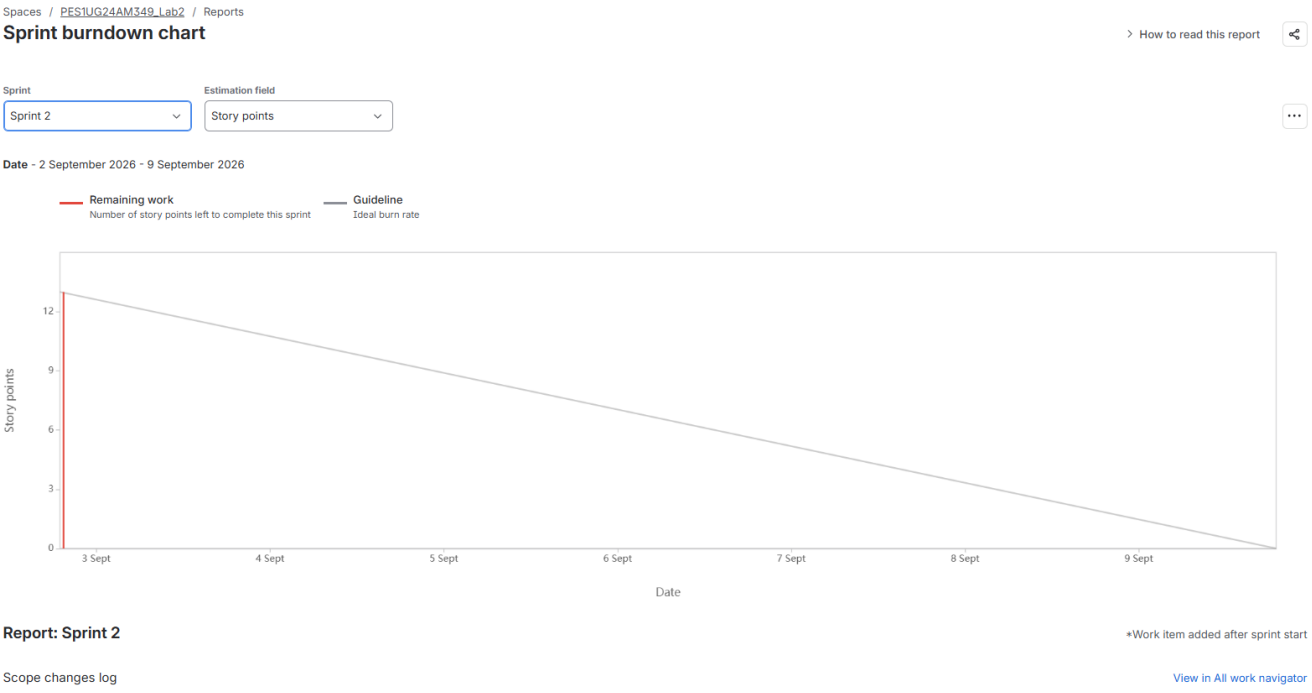


Figure 4: Sprint 2 Burndown Chart

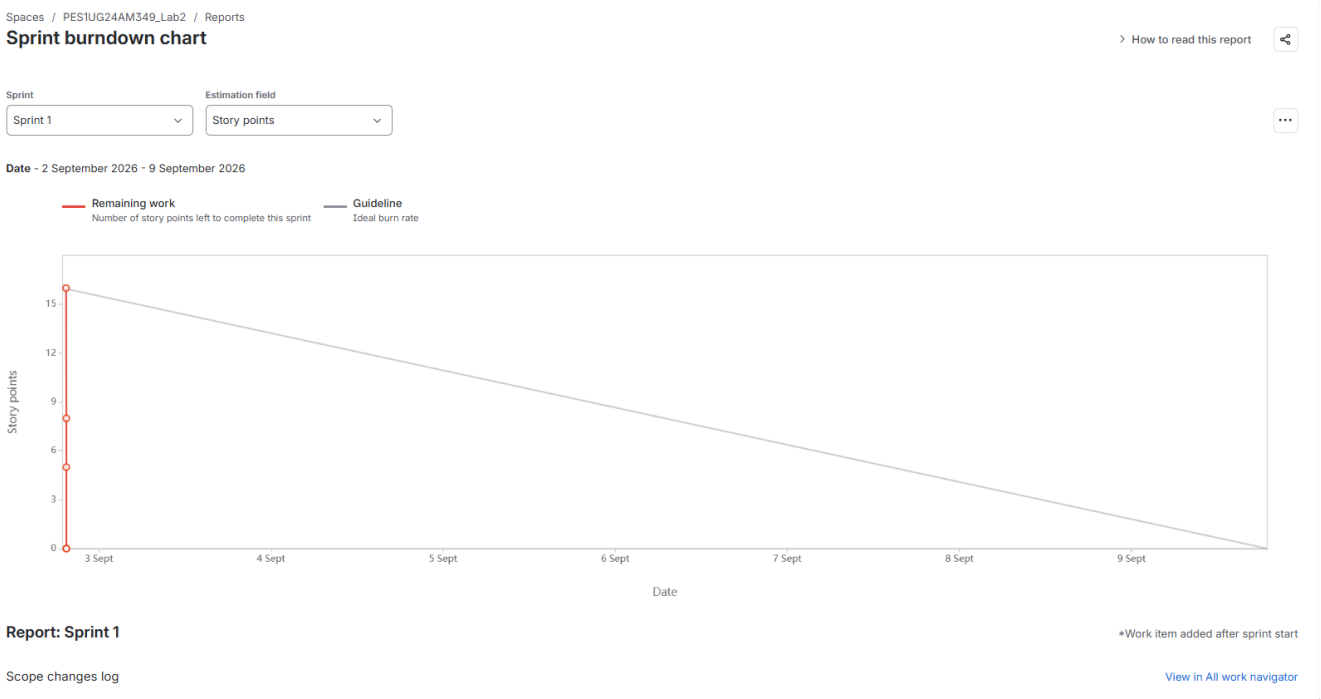


Figure 5: Sprint 1 Burndown Chart

7. Burndown Chart Analysis

The Sprint 1 Burndown Chart starts at 16 Story Points and reaches 0 as the three planned stories are completed. The Sprint 2 Burndown Chart starts at 13 Story Points and reaches 0 after the remaining three stories are completed. The charts therefore show that the planned work for both simulated sprints was completed.

8. Reflection Questions

1. Did your estimations reflect the actual effort?

Yes. The Fibonacci Story Point estimates provided a reasonable representation of the relative effort and complexity of the User Stories. Higher-point stories represented more functionality and complexity, while smaller stories represented comparatively less effort.

2. Was your backlog well-prioritized?

Yes. High-priority stories related to webhook ingestion, logging, retry handling, and failure management were prioritized because they support the core functionality of the system. Configuration and monitoring stories were assigned medium priority.

3. How did your simulated sprint align with your plan?

The simulated sprints aligned with the plan. Sprint 1 focused on core webhook ingestion, logging, and retry functionality, while Sprint 2 handled dead-letter queue management, configuration, and monitoring. All planned User Stories were progressed from To Do through In Progress to Done.

4. What insights did the burndown chart give about your team's capacity?

The Burndown Charts showed that the planned Story Points were completed within each simulated sprint. Sprint 1 completed 16 Story Points and Sprint 2 completed 13 Story Points, indicating that the selected workload was manageable for the simulated sprint capacity.

9. Conclusion

The Lab 2 exercise converted the Lab 1 functional requirements into a structured Agile backlog using Epics and User Stories. The backlog was prioritized and estimated using Fibonacci Story Points, followed by two simulated one-week sprints. The Jira Burndown Charts provided a visual representation of remaining work and confirmed completion of the planned sprint scope.